

# Application of the renormalization group to phase transitions in disordered systems\*

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The critical behavior of spin systems with quenched disorder is studied by renormalization-group methods. For the randomly dilute  $m$ -vector model, the  $n = 0$  limit is used to construct a translationally invariant effective Hamiltonian which describes the original disordered system. This Hamiltonian is analyzed in the  $\epsilon$  expansion to order  $\epsilon^2$ . Sharp second-order phase transitions with exponents which do not depend continuously on impurity concentration are predicted. For  $m > m_c \equiv 4 - 4\epsilon + O(\epsilon^2)$  the isotropic  $m$ -component fixed point, which characterizes the critical behavior of the pure system, is stable. For  $m < m_c$ , a new random fixed point becomes stable. The exponents corresponding to this fixed point are  $\eta = [(5m^2 - 8m)/256(m-1)^2]\epsilon^2 + O(\epsilon^3)$ ,  $\nu = 1/2 + [3m/32(m-1)]\epsilon + [m(127m^2 - 572m - 32)/4096(m-1)^3]\epsilon^2 + O(\epsilon^3)$  for  $m \neq 1$ , and  $\eta = -\epsilon/106 + O(\epsilon^{3/2})$ ,  $\nu = 1/2 + (6\epsilon/53)^{1/2}/4 + O(\epsilon)$  for  $m = 1$ . More general random systems are qualitatively discussed from the effective-Hamiltonian viewpoint.

## I. INTRODUCTION

The nature of phase transitions in systems with quenched or "frozen" impurities<sup>1</sup> is not yet well understood. The few rigorous results which do exist suggest that the transition in such systems may be qualitatively quite different from that in pure systems. The most compelling piece of evidence in this direction is the exact solution of a disordered two-dimensional Ising model by McCoy and Wu.<sup>2</sup> The vertical bonds in their model are taken to be random variables, constrained so that all vertical bonds in any given horizontal row are identical. In other words, the "impurities" in this model are correlated over the entire length of the crystal in the horizontal direction. The most striking feature of the model is that it displays "rounding" of the phase transition. For example, the specific heat is a nonanalytic but infinitely differentiable function of temperature at  $T_c$ , in marked contrast to the power-law behavior which occurs in typical pure systems. Unfortunately, it is difficult to ascertain whether the rounding is specific to systems wherein the impurities exhibit long-range correlations, or whether it is a general feature of disordered systems.

Another indication that critical phenomena in random systems can differ significantly from pure-system behavior has been uncovered by Griffiths.<sup>3</sup> For the randomly dilute Ising model he has demonstrated the existence of singularities in the magnetization as a function of magnetic field  $h$  at  $h=0$ , above the temperature where spontaneous magnetization occurs. While subsequent work<sup>4</sup> has helped to clarify the nature of these singular-

ities in simple dilute models, the possibility of their existence in more general random systems has yet to be explored. Whether the Griffiths singularities are symptomatic of new pathologies in the phase transition itself remains to be seen.

There are several other questions which a study of critical phenomena in the presence of quenched disorder should address. If indeed the transition for some class of random systems is sharp and second order, do the critical exponents depend upon impurity concentration?<sup>5</sup> Does the critical behavior depend upon the detailed nature of the impurities present? Perhaps most importantly, can one characterize the conditions under which disordered systems undergo sharp phase transitions and account for the rounding of the transition which is commonly observed experimentally?<sup>6</sup>

Recently, Harris<sup>7</sup> has given a very plausible heuristic answer to this last question. His argument suggests that the addition of impurities to a system which undergoes a second-order phase transition will not affect the sharpness of the transition if the specific-heat exponent  $\alpha$  of the system is negative. If  $\alpha$  is positive, Harris argues, the transition may be altered in some unspecified manner. The calculations we shall describe in this paper tend to confirm his insight.

Our approach to the problem of phase transitions in disordered systems is to transform the random problem into a translationally invariant one. We construct a translationally invariant "effective" Hamiltonian whose critical behavior is formally identical to that of the original disordered model. This approach has two benefits. First, by examining the effective Hamiltonian  $\mathcal{H}$  and comparing

it with Hamiltonians used to describe typical pure systems, one can readily deduce some of the significant features which are unique to the random problem. Second, the translational invariance of  $\mathcal{K}$  makes it directly amenable to renormalization-group analysis.<sup>8</sup> One can appeal immediately to well-established techniques and results (notably those of Brézin *et al.*<sup>9</sup>) in  $4 - \epsilon$  dimensions to arrive at conclusions about the critical behavior of the random model.

The system we shall deal with is a randomly dilute model of discrete  $m$ -component spins, interacting with short-ranged ferromagnetic forces. Making use of a result of Emery,<sup>10</sup> we point out how our calculations and results can be applied to quite arbitrary random systems, provided only that the random variables characterizing the disorder exhibit no long-range correlations. In order to obtain the effective Hamiltonian, we first transform the disordered model to the familiar continuous-spin<sup>11</sup> representation. By averaging a spin correlation function of the dilute continuous-spin system with respect to the random variables characterizing the dilution, we explicitly construct  $\mathcal{K}$ .

Having obtained  $\mathcal{K}$  we can immediately reach several conclusions about its critical behavior. The striking qualitative similarity between  $\mathcal{K}$  (which consists of exclusively short-ranged interactions) and typical pure-system Hamiltonians makes us suspect that only sharp phase transitions will occur. Since the concentration  $p$  appears in  $\mathcal{K}$  only through the functional dependence of the coefficients of the various interaction terms, we can appeal to universality and conclude that there will be no continuous variation of the critical exponents with  $p$ .

Harris and Lubensky have independently undertaken a slightly different renormalization-group approach to the random problem. Our technique involves first removing the disorder by averaging, and then employing the renormalization group. Theirs involves applying the renormalization-group transformation to the disordered system directly, and monitoring the effect of the transformation on the randomness. Using the position-space renormalization-group techniques of Niemeyer and Van Leeuwen,<sup>12</sup> they find the Ising fixed point to be stable in two dimensions.<sup>13</sup> Considering continuous-spin systems in  $4 - \epsilon$  dimensions,<sup>14</sup> they find fixed points identical to those which emerge from our effective Hamiltonian.

For all values of  $m$  except  $m = 1$ , three nontrivial fixed points of order  $\epsilon$  occur in  $4 - \epsilon$  dimensions. The first of these is the familiar isotropic  $m$ -component fixed point which characterizes the critical behavior of the pure  $m$ -component system. While this fixed point is stable for  $m > m_c$

$\equiv 4 - 4\epsilon + O(\epsilon^2)$  it is unstable whenever  $m < m_c$ . Hence the addition of impurities to the random Ising,  $XY$ , and Heisenberg models results (for small  $\epsilon$ ) in a change in critical behavior. The second fixed point is a special case of the  $m$ -component fixed point, namely,  $m = 0$ . Although it is stable for all values of  $m$ , Lubensky<sup>15</sup> has argued that it is *inaccessible* to the real physical system. Finally, a third or "random" fixed point of  $O(\epsilon)$ , stable in the region  $1 < m < m_c$ , is found. This fixed point, which is presumed to characterize the critical behavior of the random  $XY$  and Heisenberg models in 3.99 dimensions, has exponents which have been calculated to  $O(\epsilon)$  (in a context totally unrelated to random systems) by Brézin *et al.*<sup>9</sup> We compute the  $O(\epsilon^2)$  corrections to these exponents, and obtain results<sup>16</sup> identical to those of Lubensky and Harris.<sup>14</sup>

The random Ising model,  $m = 1$ , is somewhat more complicated than the other random models. The  $O(\epsilon)$  random fixed point does not occur at all for  $m = 1$ ; hence there is no stable accessible fixed point of  $O(\epsilon)$ . This fact led Lubensky<sup>15</sup> to suggest that the transition in the random Ising system might be smeared, while we postulated<sup>17</sup> that the transition might become first order. Very recently, however, we received a preprint from Khmel'nitsky,<sup>18</sup> who, by summing parquet diagrams, obtains a sharp second-order phase transition with critical exponents of  $O(\sqrt{\epsilon})$  for the random Ising system. With this information in hand, we reexamined our renormalization-group equations and found that there is indeed a fixed point of  $O(\sqrt{\epsilon})$  when  $m = 1$ . To  $O(\sqrt{\epsilon})$  this fixed point possesses one irrelevant and one marginal fourth-order operator. We believe that inclusion of  $O(\epsilon)$  corrections will show that the marginal operator is also irrelevant and hence that the new fixed point is stable. While the exponents which we compute for this new fixed point differ numerically from those found by Khmel'nitsky, this discrepancy is undoubtedly the result of a trivial algebraic error. Lubensky<sup>19</sup> has also reexamined his renormalization-group equations; he has found the  $O(\sqrt{\epsilon})$  fixed point and exponent values which agree with the results of our calculations. He now feels, as do we, that Khmel'nitsky's fixed point does characterize the critical behavior of the random Ising model. Earlier speculations about smeared or first-order transitions were probably erroneous. The phase transition in all of the random  $m$ -vector models is thus predicted to be sharp and second order.

Section II of this paper is devoted to defining the random model and casting it in the continuous spin form. A standard perturbation-theory expansion for the impurity-averaged spin-spin cor-

relation function is developed. In Sec. III the construction of the effective Hamiltonian is carried out, and a discussion of its salient features is given. The first few terms of the Hamiltonian are written out explicitly. In Sec. IV we subject this Hamiltonian (and the reader) to renormalization-group analysis in  $4 - \epsilon$  dimensions. The fixed points of the system are determined and their stability analyzed. The new fixed point, whose occurrence is associated with the presence of disorder, is discussed. The exponents corresponding to this fixed point are computed to  $O(\epsilon^2)$ . In Sec. V we discuss how our results apply to more general disordered systems. The difference between systems whose impurities are correlated or clustered over long distances and systems with purely short-ranged, or "homogeneous," disorder is heuristically discussed. Some pessimistic remarks about the experimental observability of the random-fixed-point critical exponents are made. Section VI contains a summary of results and conclusions. We try to explain the absence of Griffiths singularities in our calculation on the basis of recent work dealing with the nature of these singularities.<sup>4</sup> Finally, two appendices fill in some of the tedious arithmetic which goes into the construction of the effective Hamiltonian.

## II. MODEL

The system we shall analyze is the generalization to  $m$ -component order parameters of a random model studied by Griffiths and Lebowitz.<sup>20</sup> Consider a regular lattice consisting of  $N$  spin sites in an arbitrary number of spatial dimensions. Each site may be either occupied by a classical,  $m$ -component spin, or empty. The probability of occupation of any given site is  $p$ , where  $p$  is a fixed number between zero and one.

This probability is taken to be completely independent of the state of occupation of all other sites on the lattice. The occupation of the  $i$ th site is specified by the random variable  $p_i$ , which has the value 1 if the site is occupied and 0 if the site is vacant. The set of  $N$  values  $\{p_i\}$  then uniquely describes the occupation state of the entire lattice. The probability  $P(\{p_i\})$  of finding the system in any state  $\{p_i\}$  which has  $r$  occupied and  $(N - r)$  unoccupied sites is

$$P(\{p_i\}) = p^r (1 - p)^{N-r}. \quad (1)$$

For each configuration  $\{p_i\}$  we write a Hamiltonian of the form

$$\beta H(\{p_i\}) = -\frac{1}{2} \sum_{ij} \sum_{\alpha=1}^m K_{ij} p_i S_i^\alpha p_j S_j^\alpha, \quad (2)$$

where, as usual,  $\beta = 1/k_B T$ ,  $K_{ij}$  is some short-ranged, translationally invariant, ferromagnetic interaction, and  $S_i^\alpha$  represents one component of the spin variable for the  $i$ th site. The partition function  $\tilde{Z}(\{p_i\})$  is given by

$$\tilde{Z}(\{p_i\}) = \int_{-\infty}^{\infty} \left( \prod_{i\alpha} dS_i^\alpha \right) \left[ \prod_i \delta \left( \sum_{\alpha} S_i^{\alpha^2} - m \right) \right] \times e^{-\beta H(\{p_i\})}, \quad (3)$$

while the "impurity" averaging is done on the free energy  $F$  of the system

$$F = -\frac{1}{\beta} \sum_{\{p_i\}} P(\{p_i\}) \ln \tilde{Z}(\{p_i\}). \quad (4)$$

For  $m = 1$  this model is a random Ising model, for  $m = 2$  a random XY model, for  $m = 3$  a random Heisenberg model, and so on. To cast the partition function in a more convenient form, we perform the same series of manipulations which Hubbard<sup>11</sup> describes for the pure Ising model. This gives rise to the expression

$$\tilde{Z}(\{p_i\}) = (2\pi)^{-N/2} \int_{-\infty}^{\infty} \left( \prod_{i\alpha} dS_i^\alpha \right) \left[ \prod_i \delta \left( \sum_{\alpha} S_i^{\alpha^2} - m \right) \right] \prod_{j\beta} d\sigma_j^\beta \exp \left( -\frac{1}{2} \sum_{j\beta} (\sigma_j^\beta)^2 - \sum_{ij\beta} \sigma_j^\beta (K^{1/2})_{ji} p_i S_i^\beta \right). \quad (5)$$

The integrals over  $S_i^\alpha$  can now be done explicitly and the logarithm of the result expanded formally in powers of  $\sigma_j^\alpha$ , to arrive at the Hamiltonian

$$H(\{\sigma_j^\alpha; p_i\}) = +\frac{1}{2} \sum_{ij\alpha} L_{ij} \sigma_i^\alpha \sigma_j^\alpha - \sum_{S=2}^{\infty} p C_{2S} \sum_j \left( \sum_{\alpha} \sigma_j^{\alpha^2} \right)^S - \sum_{S=1}^{\infty} C_{2S} \sum_j Q_j \left( \sum_{\alpha} \sigma_j^{\alpha^2} \right)^S. \quad (6)$$

The quantities  $L_{ij}$  and  $Q_i$  are defined as

$$L_{ij} = (K^{-1})_{ij} - p \delta_{ij}, \quad (7a)$$

$$Q_i = p_i - p, \quad (7b)$$

respectively, while the numerical coefficients  $C_{2S}$  are simple functions of  $m$  and  $S$ .

The partition function

$$Z(\{p_i\}) = \int_{-\infty}^{\infty} \left( \prod_{j\alpha} d\sigma_j^\alpha \right) e^{-H(\{\sigma_j^\alpha; p_i\})} \quad (8)$$

gives rise to the same critical properties as does the original partition function  $\tilde{Z}(\{p_i\})$ . Defining

the spin correlation function for the state  $\{p_i\}$  as

$$\langle \sigma_i^\alpha \sigma_j^\alpha \rangle_{\{p_i\}} = \left[ \int_{-\infty}^{\infty} \left( \prod_{i\alpha} d\sigma_i^\alpha \right) e^{-H(\{\sigma_i^\alpha; p_i\})} \langle \sigma_i^\alpha \sigma_j^\alpha \rangle \right] / Z(\{p_i\}), \quad (9)$$

we can generate a diagrammatic perturbation expansion for  $\langle \sigma_i^\alpha \sigma_j^\alpha \rangle_{\{p_i\}}$  in the familiar way.<sup>8</sup> The impurity-averaged correlation function  $\langle \sigma_i^\alpha \sigma_j^\alpha \rangle$  defined by

$$\langle \sigma_i^\alpha \sigma_j^\alpha \rangle = \sum_{\{p_i\}} P(\{p_i\}) \langle \sigma_i^\alpha \sigma_j^\alpha \rangle_{\{p_i\}} \quad (10)$$

can then be obtained, in the spirit of Abrikosov, Gorkov, and Dzyaloshinski,<sup>21</sup> by averaging each of the diagrams in the perturbation series separately. In following this procedure, we assume that the operations of impurity averaging and summing the terms of the series are perfectly interchangeable. The averaging can then be incorporated directly into the diagrammatics of the perturbation expansion. A brief review of the Feynman rules which result is given in Appendix A.

### III. EFFECTIVE HAMILTONIAN

The Hamiltonian  $H(\{\sigma_i^\alpha; p_i\})$  is a function of the classical spin variables  $\{\sigma_i^\alpha\}$  and the impurity variables  $\{p_i\}$ . We have seen that the computation of a spin correlation function from this Hamiltonian involves averaging over the  $p_i$ 's. Our goal is to construct a new Hamiltonian  $\mathcal{H}(\{\phi_i^\alpha\})$  which depends only upon a set of spin variables  $\{\phi_i^\alpha\}$  and possesses translational invariance. This can be accomplished by first computing the impurity averaged correlation function  $\langle \sigma_i^\alpha \sigma_j^\alpha \rangle$  as indicated in Eq. (10), and then defining an  $\mathcal{H}(\{\phi_i^\alpha\})$  such that  $\langle \phi_i^\alpha \phi_j^\alpha \rangle$ , the spin correlation function computed with  $\mathcal{H}(\{\phi_i^\alpha\})$ , is identical with  $\langle \sigma_i^\alpha \sigma_j^\alpha \rangle$ . In this way, we absorb the essential elements of the disorder in a translationally invariant interaction.

$H(\{\sigma_i^\alpha; p_i\})$  contains complicated interaction terms involving arbitrarily large powers of the variables  $\{\sigma_i^\alpha\}$ . Clearly  $\mathcal{H}(\{\phi_i^\alpha\})$  will also include such many-body interactions, as well as additional complications from the presence of the impurities. Some of the features which distinguish the random problem from pure systems will be evident from a cursory inspection of  $\mathcal{H}(\{\phi_i^\alpha\})$ .

The construction of  $\mathcal{H}$  can be understood through consideration of the simplest random term of  $\mathcal{H}(\{\sigma_i^\alpha; p_i\})$ , namely,  $C_2 \sum_{j\alpha} Q_j \sigma_j^{\alpha 2}$ . The lowest-order graph which results from this term and contributes to the impurity-averaged spin-spin correlation function is shown in Fig. 1. The obvious topological similarity between this diagram and the lowest order exchange diagram for two-body



FIG. 1. Lowest-order contribution to the impurity-averaged spin-spin correlation function.

potentials<sup>22</sup> suggests that the most elementary term of the effective Hamiltonian should be a four-spin interaction.

We test this hypothesis by writing

$$\mathcal{H}(\{\phi_i^\alpha\}) = \frac{1}{2} \sum_{ij\alpha} L_{ij} \phi_i^\alpha \phi_j^\alpha + \frac{1}{2} V \sum_{ij\alpha'} \delta_{ij} \phi_i^{\alpha 2} \phi_j^{\alpha' 2}. \quad (11)$$

The four-spin term of this Hamiltonian is represented graphically in Fig. 2(a). In lowest-order perturbation theory for the quantity  $\langle \phi_i^\alpha \phi_j^\alpha \rangle$ , this interaction gives rise to the diagram shown in Fig. 2(b). It is a trivial matter to compute the analytic expression corresponding to this graph and compare it with the expression arising from Fig. 1. In order to make these expressions identical, we need only define the potential strength  $V$  to be

$$V = (C_2)^2 p(1-p). \quad (12)$$

The obvious difficulty with this procedure is that the interaction (11) gives rise not only to the "exchange" diagram of Fig. 2(b), but to the "direct" graph of Fig. 3. This latter graph has no counterpart in the original impurity-averaged theory. Fortunately, it can be eliminated from the effective Hamiltonian theory by a mathematical

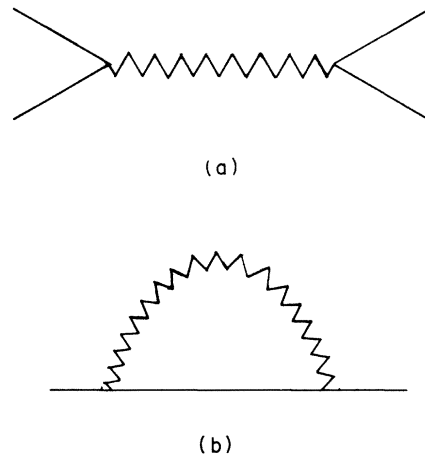


FIG. 2. (a) Simplest vertex of the effective Hamiltonian ( $\mathcal{H}$ ) theory. (b) Lowest-order "exchange" graph of the effective Hamiltonian theory.

trick, devised by de Gennes<sup>23</sup> for the excluded volume problem. The idea is to affix an extra component label to the order parameter, thus producing the new order parameter  $\phi_{i\beta}^\alpha$ . The new component label is to be summed from 1 to  $n$ , so that the order parameter now has  $mn$  components. The effective Hamiltonian can then be rewritten

$$\mathcal{H}(\{\phi_{i\beta}^\alpha\}) = \frac{1}{2} \sum_{ij} \sum_{\alpha=1}^m \sum_{\beta=1}^n L_{ij} \phi_{i\beta}^\alpha \phi_{j\beta}^\alpha + \frac{1}{2} V \sum_{ij} \sum_{\alpha, \alpha'=1}^m \sum_{\beta, \beta'=1}^n \delta_{ij} (\phi_{i\beta}^\alpha)^2 (\phi_{j\beta'}^{\alpha'})^2, \quad (13)$$

and the correlation function  $\langle \phi_{i\beta}^\alpha \phi_{j\beta}^\alpha \rangle$  defined in the obvious way. The lowest-order graphs for this quantity have the topologies of Figs. 2(b) and 3. The analytic expression corresponding to Fig. 2(b) is unaltered by the presence of the extra component label. However, the direct graph of Fig. 3 acquires an extra multiplicative factor of  $n$ , because of the increased number of components. Setting  $n=0$  automatically eliminates this extraneous diagram from the theory. In similar fashion, the  $n=0$  constraint removes unwanted graphs of all higher orders from the perturbation expansion. It should be understood as a calculational prescription: compute correlation functions, critical exponents, etc., as a function of  $n$ ,

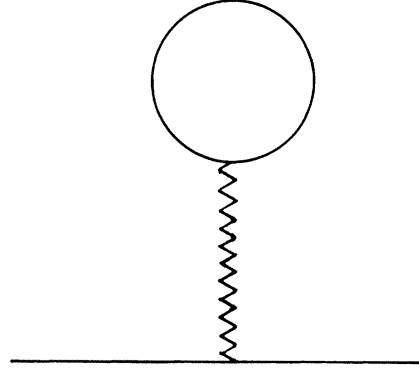


FIG. 3. Lowest-order "direct" graph of the effective Hamiltonian theory.

and set  $n=0$  at the end of the calculation.

The lowest-order terms of the effective Hamiltonian have been written in Eq. (13). The construction of all higher-order terms proceeds in similar fashion and involves nothing new in principle. The details of this process are straightforward but lengthy, and are given in Appendix B, where the effective Hamiltonian is exhibited in full. It is useful for later purposes to consider a truncated form, which retains all terms up to order  $\phi^6$ . This is given by

$$\begin{aligned} \mathcal{H}(\{\phi_{i\beta}^\alpha\}) = & \frac{1}{2} \sum_{ij} L_{ij} \phi_{i\beta}^\alpha \phi_{j\beta}^\alpha - p C_4 \sum_{j\alpha\alpha'} (\phi_{j\beta}^\alpha)^2 (\phi_{j\beta'}^{\alpha'})^2 - \frac{C_2^2}{2!} p(1-p) \sum_{j\alpha\alpha'} (\phi_{j\beta}^\alpha)^2 (\phi_{j\beta'}^{\alpha'})^2 - p C_6 \sum_{j\alpha\alpha'\alpha''} (\phi_{j\beta}^\alpha)^2 (\phi_{j\beta'}^{\alpha'})^2 (\phi_{j\beta''}^{\alpha''})^2 \\ & - \frac{C_3^2}{3!} p(1-p)(1-2p) \sum_{j\alpha\alpha'\alpha''} (\phi_{j\beta}^\alpha)^2 (\phi_{j\beta'}^{\alpha'})^2 (\phi_{j\beta''}^{\alpha''})^2 - C_2 C_4 p(1-p) \sum_{j\alpha\alpha'\alpha''} (\phi_{j\beta}^\alpha)^2 (\phi_{j\beta'}^{\alpha'})^2 (\phi_{j\beta''}^{\alpha''})^2 \end{aligned} \quad (14)$$

and provides the basis for our analysis of the critical behavior of the random system.

Recently, Emery<sup>10</sup> has developed an alternate method of obtaining the effective Hamiltonian. He has shown that, for systems with the  $n=0$  constraint, averaging the free energy with respect to random (impurity) variables is perfectly equivalent to averaging the partition function. This fact enables one to derive  $\mathcal{H}(\{\phi_{i\beta}^\alpha\})$  directly from the partition function of Eq. (8), without using perturbation theory. The effective Hamiltonian obtained by Emery's methods can be shown to be identical to the one constructed here.

The most striking feature of  $\mathcal{H}(\{\phi_{i\beta}^\alpha\})$  is that, although it consists of many-body interactions of arbitrarily high order, each of these terms is explicitly short ranged. In fact, only site diagonal, or point, interactions occur. This property is readily accounted for on physical grounds. The

interaction terms of the effective Hamiltonian represent the effect of the impurities in the system. The range of these effective interactions is governed by the characteristic length over which the impurities at different sites on the lattice are correlated. The impurities in the present model are, of course, totally uncorrelated. [The mathematical statement of this fact is Eq. (A2)]. The site diagonal nature of the effective interactions results from this absence of correlation.

The fact that the dilute models possess translationally invariant effective Hamiltonians with exclusively short-range forces has immediate consequences. The only traces of disorder which remain in the effective Hamiltonians are the  $n=0$  constraint and the dependence of the coefficients of the various interaction terms on concentration  $p$ . For all intents and purposes, therefore, these Hamiltonians are indistinguishable from those

typically used to consider phase transitions in pure systems. Before doing any further calculation we are led to the conclusion that no rounded transitions will result from the effective Hamiltonians. For if rounding or other pathological behavior does arise, then surely it must arise in pure systems as well. Of course it is conceivable that the simple analytic continuation to  $n=0$  is too naive and fails to detect qualitatively new critical phenomena which set in somewhere below  $n=1$ . However, the critical exponents which emerge from renormalization-group treatments do have (as we shall see) smooth, well defined  $n=0$  limits whose existence argues against this possibility. Thus our prejudice is that effective Hamiltonians

of the type we have obtained give rise to sharp, "orthodox" phase transitions. Later on we shall speculate about the effect of long-range forces in effective Hamiltonians.

#### IV. FIXED-POINT ANALYSIS

It is natural to apply standard renormalization-group methods to investigate the critical behavior of  $\mathcal{H}(\{\phi_{i\beta}\})$ . In  $4-\epsilon$  dimensions, only interaction terms of order  $\phi^4$  or less have any relevance<sup>8</sup> for determining the fixed points of the system. In order to proceed with the renormalization-group analysis, the continuum representation of  $\mathcal{H}$  is the most convenient to study. This can be written

$$\mathcal{H} = \int d^d x \left( \frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} m_0^2 \phi^2(x) + u_0 \Lambda^\epsilon \sum_{\substack{\alpha \alpha' \\ \beta \neq \beta'}} [\phi_\beta^\alpha(x)]^2 [\phi_{\beta'}^{\alpha'}(x)]^2 + w_0 \Lambda^\epsilon \sum_{\substack{\alpha \alpha' \\ \beta}} [\phi_\beta^\alpha(x)]^2 [\phi_\beta^{\alpha'}(x)]^2 \right), \quad (15)$$

where  $u_0$  and  $w_0$  are dimensionless parameters,  $\Lambda$  is the reciprocal-lattice constant, and terms of order  $\phi^6$  and higher have been omitted. The quadratic terms are defined by

$$(\nabla \phi)^2 = \sum_{\alpha=1}^m \sum_{\beta=1}^n [\nabla \phi_\beta^\alpha(x)]^2, \quad (16a)$$

$$\phi^2(x) = \sum_{\alpha, \beta} [\phi_\beta^\alpha(x)]^2. \quad (16b)$$

Note that the  $w_0$  interaction term in the Hamiltonian consists of all  $\phi^4$  terms with four identical lower component labels. The  $u_0$  interaction contains no such terms. This grouping of the  $\phi^4$  terms proves convenient for later calculations.

A useful method for extracting the fixed points of Hamiltonian (15) is the technique of Brézin *et al.*,<sup>24</sup> which utilizes renormalized perturbation theory and the Callan-Symanzik<sup>25</sup> equations in  $(4-\epsilon)$  dimensions. We perform mass, wave-function, and coupling-constant renormalizations in the conventional fashion,

$$\begin{aligned} \mathcal{H} = \int d^d x \left( Z_3 \left[ \frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} m^2 \phi^2(x) \right] + \frac{um^\epsilon Z_1}{4!} \right. \\ \left. \times \sum_{\substack{\alpha \alpha' \\ \beta \neq \beta'}} \phi_\beta^{\alpha^2} \phi_{\beta'}^{\alpha'^2} + \frac{wm^\epsilon Z_2}{4!} \sum_{\substack{\alpha \alpha' \\ \beta}} \phi_\beta^{\alpha^2} \phi_\beta^{\alpha'^2} + \frac{1}{2} \delta m^2 \phi^2(x) \right), \end{aligned} \quad (17)$$

using the zero-momenta renormalization conditions

$$\Gamma_{\beta\beta}^{(2)\alpha\alpha}(p, m, u, w)|_{p^2=0} = m^2, \quad (18a)$$

$$\frac{d}{dp^2} \Gamma_{\beta\beta}^{(2)\alpha\alpha} \Big|_{p^2=0} = 1, \quad (18b)$$

$$\Gamma_{\beta\beta \neq \beta' \beta'}^{(4)\alpha\alpha\alpha\alpha} \Big|_{p=0} = \frac{1}{3} um^\epsilon, \quad (18c)$$

$$\Gamma_{\beta\beta\beta\beta}^{(4)\alpha\alpha\alpha\alpha} \Big|_{p=0} = wm^\epsilon. \quad (18d)$$

The symbol  $\Gamma^{(r)}$  represents the  $r$ -point, amputated, single-particle irreducible vertex function. The components of the external legs of the vertex are indicated by subscripts and superscripts attached to  $\Gamma^{(r)}$ .

Vertex functions whose momenta  $p_i$  satisfy  $p_i \gg m$  have asymptotic forms which can be computed through use of the homogeneous Callan-Symanzik equations

$$\left( m \frac{\partial}{\partial m} + \beta_u(uw) \frac{\partial}{\partial u} + \beta_w(uw) \frac{\partial}{\partial w} - \frac{r}{2} \gamma_3 \right) \Gamma^{(r)} = 0. \quad (19)$$

The  $\beta$  functions and  $\gamma_3$  are defined by

$$\gamma_3 = m \left( \frac{\partial \ln Z_3}{\partial m} \right) \Big|_{w_0, u_0}, \quad (20a)$$

$$\beta_u(uw) = m \left( \frac{\partial u}{\partial m} \right) \Big|_{w_0, u_0}, \quad (20b)$$

$$\beta_w(uw) = m \left( \frac{\partial w}{\partial m} \right) \Big|_{w_0, u_0}. \quad (20c)$$

Introduction of the dimensionless parameter  $l$ , defined by

$$l = \ln(m/\Lambda), \quad (21)$$

gives these last two equations a form exactly analogous to the differential form of Wilson's renor-

malization-group equations,<sup>26</sup> namely,

$$\frac{\partial u}{\partial l} = \beta_u, \quad (22a)$$

$$\frac{\partial w}{\partial l} = \beta_w. \quad (22b)$$

The renormalized mass  $m$  is simply the inverse of the correlation length  $\xi$ .<sup>24</sup> Thus the limit  $m \rightarrow 0$  (or equivalently  $l \rightarrow -\infty$ ) is the appropriate limit for discussing critical behavior. It is clear from Eqs. (22) that, as  $l \rightarrow -\infty$ , the renormalized coupling constants  $u$  and  $w$  may be driven toward values  $u^*$  and  $w^*$  which are simultaneous zeros of the  $\beta$  functions. Such values are the fixed points of the system. The functions  $\beta_u$  and  $\beta_w$  are easily calculable as power series in  $u$  and  $w$  for arbitrary values of  $m$  and  $n$ . To cubic order, they are

$$\begin{aligned} \beta_w = & -\epsilon w + \frac{1}{6}(m+8)(1 - \frac{1}{2}\epsilon)Sw^2 \\ & + \frac{1}{6}m(n-1)(1 - \frac{1}{2}\epsilon)Su^2 - \frac{5}{36}m(n-1)S^2u^2w \\ & - \frac{1}{9}m(n-1)S^2u^3 - \frac{1}{12}(3m+14)S^2w^3, \end{aligned} \quad (23a)$$

$$\begin{aligned} \beta_u = & -\epsilon u + \frac{1}{3}(m+2)(1 - \frac{1}{2}\epsilon)Suw + \frac{1}{6}[m(n-2)+4] \\ & \times (1 - \frac{1}{2}\epsilon)Su^2 + \frac{1}{36}(-9mn+17m-8)S^2u^3 \\ & - \frac{1}{3}(m+2)S^2u^2w - \frac{5}{36}(m+2)S^2uw^2, \end{aligned} \quad (23b)$$

where  $S$  is a numerical factor given by<sup>24</sup>

$$S = 2\pi^{d/2}/(2\pi)^d \Gamma(\frac{1}{2}d). \quad (24)$$

Brézin, Le Guillou, and Zinn-Justin<sup>9</sup> have studied the fixed points of this system, to  $O(\epsilon)$ , for positive integral values of  $m$  and  $n$ . They find only one stable fixed point corresponding to each  $(m, n)$  pair. For purposes of the impurity problem, however, we are interested in the case  $n=0$ . In general there are four fixed points for this value of  $n$  when  $\epsilon$  is small. They are listed in Table I. The first of the three nontrivial fixed points given in this table is the usual isotropic  $m$ -vector

model fixed point, characterized by the familiar critical exponents which Wilson<sup>27</sup> has calculated. The second is a special case of the  $m$ -vector fixed point, namely,  $m=0$ . Since this fixed point arises because of the  $n=0$  constraint, we refer to it as the " $n=0$ " fixed point. The third, or "random" fixed point gives rise to a less familiar set of critical exponents. In order to discuss this fixed point, it is convenient to consider the cases  $m \neq 1$  and  $m=1$  separately. For  $m \neq 1$ , the critical exponents of the random fixed point have been calculated to  $O(\epsilon)$  by Brézin *et al.*<sup>9</sup> The values of  $\eta$  and  $\nu$  to  $O(\epsilon^2)$  are listed in Table II, the other exponents being readily obtainable from scaling laws.<sup>28</sup> For the random Ising system,  $m=1$ , no random fixed point of  $O(\epsilon)$  can be found. (We are alerted to this fact by the blowing up of the critical exponents in Table II when  $m=1$ .) However, as Khmel'nitsky<sup>18</sup> has pointed out, there is a fixed point of  $O(\sqrt{\epsilon})$  in this case. The critical exponents are given to  $O(\sqrt{\epsilon})$  in Table III.

The last column in Table I shows that the isotropic  $m$ -component fixed point is unstable when  $m$  is less than a critical value  $m_c$ , given by

$$m_c = 4(1 - \epsilon) + O(\epsilon^2). \quad (25)$$

For such values of  $m$  the addition of impurities to the lattice drives the system away from the pure ( $m$ -component) fixed point in 3.99 dimensions. It is the presence of the  $u_0$  term in Hamiltonian (15) which produces the instability. It is straightforward to calculate the crossover exponent  $\phi_m$  which characterizes the departure from the pure fixed point. A convenient method for so doing is the technique of Brézin, de Dominicis, and Zinn-Justin.<sup>29</sup> The result is

$$\phi_m = \frac{(4-m)}{2(m+8)}\epsilon - \frac{(m+2)^2(m+28)}{4(m+8)^3}\epsilon^2 + O(\epsilon^3). \quad (26)$$

TABLE I. Fixed points.

| Fixed point       | $Su^*$                                                                                           | $Sw^*$                                                                                                    | Region of stability                              |
|-------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Gaussian          | 0                                                                                                | 0                                                                                                         | Never                                            |
| $m$ component     | 0                                                                                                | $\frac{6\epsilon}{m+8} \left[ 1 + \epsilon \left( \frac{1}{2} + \frac{3(3m+14)}{(m+8)^2} \right) \right]$ | $m > m_c \equiv 4(1 - \epsilon) + O(\epsilon^2)$ |
| $n=0$             | $\frac{3}{4}\epsilon + \frac{111}{128}\epsilon^2$                                                | $\frac{3}{4}\epsilon + \frac{111}{128}\epsilon^2$                                                         | All $m$                                          |
| Random $m \neq 1$ | $\frac{3(m-4)\epsilon}{8(m-1)} + \frac{3(169m^3 - 748m^2 + 1568m - 512)\epsilon^2}{1024(m-1)^3}$ | $\frac{3m\epsilon}{8(m-1)} + \frac{m(507m^2 - 1176m + 192)\epsilon^2}{1024(m-1)^3}$                       | $m < m_c \equiv 4(1 - \epsilon) + O(\epsilon^2)$ |
| Random $m=1$      | $-(\frac{54}{53})^{1/2}\sqrt{\epsilon} + O(\epsilon)$                                            | $(\frac{6}{53})^{1/2}\sqrt{\epsilon} + O(\epsilon)$                                                       | ?                                                |

TABLE II. Exponents of the random fixed point for  $m \neq 1$ .

| Exponent | Random-fixed-point value                                                                      |
|----------|-----------------------------------------------------------------------------------------------|
| $\eta$   | $\frac{m(5m-8)\epsilon^2}{256(m-1)^2}$                                                        |
| $\nu$    | $\frac{1}{2} + \frac{3m\epsilon}{32(m-1)} + \frac{m(127m^2-572m-32)}{4096(m-1)^3} \epsilon^2$ |

The  $m$ -component fixed point is stable for  $m > m_c$  as is indicated by the change of sign which  $\phi_m$  undergoes at  $m = m_c$ . Subsequent to the completion of this work, Aharony<sup>30</sup> proved that  $\phi_m$  is identically equal to  $\alpha$ , the specific-heat exponent of the isotropic  $m$ -component fixed point. His proof uses very general renormalization-group arguments and is valid for any dimension. The critical behavior of pure systems with positive values of  $\alpha$  will therefore be modified in the presence of disorder.

It is trivial to verify that, to  $O(\epsilon^2)$ , the value of  $\alpha$  for the random fixed point passes through zero when  $m$  attains the critical value  $m_c$ . Hence there is a single accessible stable fixed point for all values of  $m$  near four dimensions; this stable fixed point is characterized by a negative value of  $\alpha$ . These results are in perfect agreement with Harris's<sup>7</sup> heuristic argument which predicts that phase transitions with positive values of  $\alpha$  do not occur in the presence of quenched disorder.

The stability of Khmel'nitsky's random fixed point for  $m = 1$  is in some doubt. To  $O(\sqrt{\epsilon})$  this fixed point possesses one irrelevant fourth-order operator and one marginal fourth-order operator. While we believe that extending the stability calculations to  $O(\epsilon)$  will show the marginal operator to be irrelevant and hence prove the stability of the fixed point, such calculations are extremely tedious and have not yet been carried out. This uncertainty has been indicated by the question mark in the last column of Table I.

It is worth emphasizing that the stability we have been discussing is purely a local property of the fixed point. While the existence of a unique accessible stable fixed point for each value of  $m$  strongly suggests that this fixed point governs the

critical behavior of the disordered system, we cannot prove it. Given some arbitrary value of the concentration  $p$  we cannot establish that the system evolves to the stable fixed point under renormalization-group transformation. It is entirely possible that critical behavior more complex than a simple second-order phase transition could occur. We conclude this section with a heuristic discussion of one such possibility, namely, the occurrence of a first-order phase transition.

For simplicity we restrict ourselves to the dilute Ising model. The first few terms of the effective Hamiltonian for this system can be written

$$\mathcal{H} = \int d^d x \left[ \frac{1}{2} \sum_{\alpha=1}^n [(\nabla \phi_{\alpha})^2 + r_0 \phi_{\alpha}^2] + w_0 \sum_{\alpha=1}^n \phi_{\alpha}^4 + u_0 \left( \sum_{\alpha=1}^n \phi_{\alpha}^2 \right)^2 + v_0 \left( \sum_{\alpha=1}^n \phi_{\alpha}^2 \right)^3 \right]. \quad (27)$$

We recall from Eq. (14) that the coefficient  $u_0$  is negative for the random system. The  $(\phi^2)^3$  term, positive for  $p > \frac{1}{2}$ , has been included to ensure the stability of the Hamiltonian. The presence of negative  $\phi^4$  and positive  $\phi^6$  interactions is immediately reminiscent of the continuum Blume-Emery-Griffiths<sup>31</sup> (BEG) model analyzed by Riedel and Wegner.<sup>32</sup> Hence we are alerted to the possible occurrence of tricritical behavior and first-order phase transitions. Let us briefly review how these phenomena come about in Riedel and Wegner's analysis. The  $n$ -component BEG model is described by the Hamiltonian (27) with  $w_0$  set equal to zero. In the limit  $n=0$ , this Hamiltonian possesses two fixed points: the stable  $n=0$  fixed point and the unstable Gaussian fixed point. Now suppose  $v_0$  is fixed at some arbitrary positive value, and imagine decreasing  $u_0$  toward zero through positive values. As long as  $u_0$  is positive, the system is in the regime of the critical line; it undergoes second-order phase transitions with  $n=0$  critical exponents. This critical line persists as  $u_0$  passes through zero. As  $u_0$  continues to decrease, however, a value of  $u_0$  will be reached where the competing effects of the negative  $\phi^4$  and positive  $\phi^6$  terms will cancel. This point is the tricritical point, and is characterized by Gaussian exponents. Decreasing  $u_0$  further allows the negative  $\phi^4$  term to dominate and puts us in the region of first-order phase transitions. The familiar tricritical phase diagram results.

The presence of a nonvanishing  $w_0$  term (i.e., a term with cubic symmetry) complicates the situation considerably. Two more fixed points, the (presumably) stable random and the unstable Ising, occur. Note from Table I that the  $n=0$  and random fixed points have positive and negative values of  $u^*$ , respectively. (Recall that the start-

TABLE III. Exponents of the random fixed point for  $m = 1$ .

| Exponent | Random-fixed-point value                                                                    |
|----------|---------------------------------------------------------------------------------------------|
| $\eta$   | $-\frac{1}{106} \epsilon + O(\epsilon^{3/2})$                                               |
| $\nu$    | $\frac{1}{2} + \frac{1}{4} \left( \frac{5}{53} \right)^{1/2} \sqrt{\epsilon} + O(\epsilon)$ |

ing coefficient  $u_0$  for the physical random Ising system is negative. It was by showing that this coefficient cannot change sign under renormalization-group transformation that Lubensky and Harris<sup>14</sup> proved the inaccessibility of the  $n=0$  fixed point to the true physical system.) Now imagine fixing  $w_0$  and  $v_0$  at positive values and decreasing  $u_0$  toward zero.<sup>33</sup> The system presumably has a line of  $n=0$  second-order phase transitions for positive  $u_0$ ; the line persists down through  $u_0=0$ . Eventually  $u_0$  will reach the "tricritical" value where the effects of the  $u_0$  and  $v_0$  terms cancel. At this point, because of the presence of the  $w_0$  term, the system will undergo a second-order transition with Ising critical exponents. As  $u_0$  is decreased still further, we move toward the physical region. It is here that the speculation starts. The most plausible assumption by far is that a line of second-order transitions with random fixed-point exponents begins. If this is indeed the case then, as postulated earlier, the random fixed point does characterize the critical behavior of the physical random system. However, it is as yet impossible to rigorously discount the possibility that the transition turns first order at the "Ising tricritical" point. In this eventuality, the physical random system would undergo a first-order phase transition.

The foregoing discussion has been highly heuristic and is intended merely to raise some of the global questions that fixed-point analysis does not resolve. A much more detailed treatment of the problem is needed in order to put to rest for once and for all the spectre of first-order transitions. Very recently, Aharony, Imry, and Ma<sup>34</sup> have investigated the possibility of first-order transitions in the random Gaussian model—the model of Eq. (27) with  $w_0$  equal to zero. They have stressed the importance of considering not only the  $\phi^4$  and  $\phi^6$  terms but all of the higher-order terms in the Hamiltonian as well. The effect of these higher terms can be to prevent the formation of the usual  $\phi^4$ - $\phi^6$  potential well and hence to suppress the first-order transition. Indeed, these workers see no evidence of a first-order transition in the random Gaussian model when the high-order interactions are considered. Similar careful analysis for the random Ising model is clearly required.

#### V. MORE GENERAL DISORDERED SYSTEMS: EXPERIMENTAL RAMIFICATIONS

The effective Hamiltonian technique has rather broad implications for the general question of the nature of phase transitions in disordered systems. We have seen that we could (formally at least) represent the dilute  $m$ -vector model by a trans-

lationally invariant Hamiltonian wherein effective interactions and the  $n=0$  constraint accounted for the presence of disorder. The crucial feature of these effective interactions was their short range. We argued that this property was a manifestation of the site diagonal nature of the randomness in the model. This observation suggests that we should classify systems with quenched disorder as belonging in one of two categories. Systems wherein the random variables exhibit no long-range correlations constitute the first class. It is clear that application of our methods to systems of this class will result in Hamiltonians consisting of purely short-ranged interactions.<sup>35</sup> The second class, into which the model of McCoy and Wu falls, is composed of systems wherein the random variables exhibit long-range correlations. Effective Hamiltonians corresponding to members of this class will contain long-range forces.

We expect the effective Hamiltonians of systems of the first class to be virtually indistinguishable from pure-system Hamiltonians. Barring total failure of the analytic continuation to  $n=0$ , critical behavior (such as rounding) not found in pure systems will therefore not occur for members of this class. Possible changes in the values of critical exponents and perhaps even a change in the order of the transition are the most drastic effects of disorder that we anticipate. No continuous variation of exponents with concentration will result.

Systems of the second class pose a more complex problem. The calculations of McCoy and Wu indicate that rounding of the transition can occur. Lubensky<sup>15</sup> has shown that the generalization of the McCoy-Wu model to  $4-\epsilon$  dimensions possesses no stable fixed point. It is very appealing to infer that the absence of a stable fixed point in such disordered systems is symptomatic of a smeared transition. However until one understands how to explicitly detect rounding of the transition through renormalization group methods, one can only speculate as to how general a phenomenon rounding is and whether or not the details of the smearing are dependent upon the precise nature of the disorder.

It is clearly of great interest to try to observe the "random" critical exponents experimentally. Recall that the exponent  $\phi$  characterizing the crossover from the pure to the random fixed point is identically equal to  $\alpha$ , the pure-system specific-heat exponent. Thus only pure systems having  $\alpha > 0$  could be expected to exhibit crossover to the random fixed point. According to available experimental information only Ising systems in three dimensions fulfill this criterion.<sup>36</sup> Unfortunately,  $\alpha$  is approximately<sup>36</sup>  $\frac{1}{8}$  for such systems. This

is a rather small value and will make the crossover region very narrow and hard to observe.<sup>37</sup> Furthermore, the exponents of the random fixed point are known only to  $O(\sqrt{\epsilon})$  for the Ising model. While we don't know the optimum place to truncate a  $\sqrt{\epsilon}$  series, it is likely that at least terms of  $O(\epsilon)$  should be retained in order to give good results in three dimensions. Thus the values of the random critical exponents are highly uncertain. It is possible that the correct values are so close to the pure ones as to be indistinguishable experimentally.

Our calculations indicate that the phase transition in a broad class of quenched disordered systems is sharp and second order. How then does one account for the rounding which is observed in so many experiments in critical phenomena?<sup>6</sup> This is a very complex question to which we have no convincing answer. One possibility is that the presence of macroscopic blocks or clusters of impurities in a real sample can inhibit the growth of the correlation length and give rise to rounding. However it is unclear that one can account for all experimental smearing on the basis of such inhomogeneous disorder. Any experiments wherein the characteristic size of the impurity clusters could be varied in a systematic manner would be an invaluable aid to the solution of this problem.

## VI. SUMMARY AND CONCLUSIONS

In this paper we have shown, through use of the  $n=0$  limit, that the problem of magnetic phase transitions in quenched disordered systems can be cast in the form of a translationally invariant problem. The translationally invariant system can then be treated straightforwardly by standard renormalization-group methods. If the randomness is homogeneous (that is, no macroscopic clustering of impurities occurs) the phase transition is predicted to be sharp. Critical exponents do not depend continuously on impurity concentration. In  $4 - \epsilon$  dimensions, homogeneously random systems possess a unique accessible stable fixed point. If  $\alpha$ , the specific-heat exponent of the corresponding pure system, is negative, then the usual isotropic "pure" fixed point is stable. If  $\alpha$  is positive, a new random fixed point, whose specific-heat exponent is negative, becomes stable.

These results confirm a heuristic argument due to Harris.<sup>7</sup> Only systems with positive  $\alpha$  are expected to display the random-fixed-point critical exponents. The crossover exponent characterizing the transition from pure to random fixed point is equal to  $\alpha$ . Hence the critical region is expected to be small and experimental observation of the random exponents will be difficult. Moreover,

any macroscopic clumping of the impurities in a sample may round the phase transition.

It is worth emphasizing that these conclusions are subject to all the usual uncertainties inherent in simple fixed-point analyses. The presence of a unique accessible stable fixed point in  $4 - \epsilon$  dimensions is strongly suggestive of second-order transitions, but until more careful studies of global flows in Hamiltonian space are performed it does not constitute a proof. Extrapolation of our results to three and two dimensions should also be carried out with due caution.

The calculations we have described are incapable of detecting the Griffiths singularities which are known to be present in the dilute Ising model. One should ask if this failure is indicative of a serious inadequacy in our analysis. While it is impossible to answer this question rigorously, we believe that at least there is nothing incompatible between the presence of Griffiths singularities and our results. Since all of our calculations were done in zero magnetic field the only indication of the singularities we could hope to see would be a blowing up of one of the derivatives of the susceptibility at  $h=0$  above  $T_c$ . However, recent calculations by Wortis and by Harris<sup>4</sup> suggest that the Griffiths singularities are sufficiently weak that all derivatives of the magnetization with respect to field are perfectly finite at  $h=0$ . If this is indeed the case, then no zero-field calculation which merely examines a few spin correlation functions could possibly detect the Griffiths non-analyticities. Moreover, the renormalization group is designed to pick out asymptotic behavior near a point where the correlation length of the system becomes infinite. Unless (as seems highly unlikely in view of Ref. 4) the Griffiths singularities involve a divergence of the susceptibility itself above  $T_c$ , standard renormalization-group methods cannot be expected to detect them. Thus the failure of the nonanalyticities to appear in our calculation need not imply that there is a fatal flaw in our procedure. Until one understands how to see these singularities with renormalization-group techniques, however, there remains the worry that forms of critical behavior more exotic than second-order transitions are simply invisible to renormalization-group calculations.

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## APPENDIX A: FEYNMAN RULES

The spin correlation function  $\langle \sigma_i^\alpha \sigma_j^\alpha \rangle_{\{p_i\}}$  and the Hamiltonian  $H(\{\sigma_i^\alpha; p_i\})$  were defined in Eqs. (9) and (6), respectively. We now expand all of the exponentials appearing in Eq. (9), except for the  $L_{ij}$  term, to generate a perturbation expansion in the usual way.<sup>8</sup> In this expansion, each factor of  $p C_{2S} \sum_j [\sum_\alpha (\sigma_j^\alpha)^2]^S$  will be referred to as a “2Sth-order vertex of type A” and can be represented graphically as the 2S-point vertex shown in Fig. 4(a). Each factor of  $C_{2S} \sum_j Q_j [\sum_\alpha (\sigma_j^\alpha)^2]^S$  will be referred to as a “2Sth-order vertex of type B” and can be represented by the 2S-point vertex shown in Fig. 4(b). The Feynman rules for computing  $\langle \sigma_i^\alpha \sigma_j^\alpha \rangle_{\{p_i\}}$  have the following structure: (a) Draw all possible topologically distinct, connected graphs, each with two external legs, consisting of arbitrary numbers of vertices (of arbitrary order) of both types A and B. The external legs bear site labels  $i$  and  $j$ ; each bears the component label  $\alpha$ . (b) Give each vertex a site label, and give each pair of lines emerging from the vertex a component label. (c) With each line joining two vertices with site labels  $i$  and  $j$ , associate the factor  $L_{ij}^{-1}$ . These “propagators” are diagonal in the component indices. (d) With each vertex of order 2S and type A, associate the factor  $p C_{2S}$ . With each ver-

tex of order 2S and type B, associate the factor  $C_{2S} Q_i$ , where  $i$  is the site label of the vertex. (e) Multiply all factors obtained from rules (c) and (d) together, and sum the product over all internal vertex labels, letting each vertex sum run over all sites of the lattice. Sum the result over all independent component labels. (f) Multiply the result of (e) by a numerical factor appropriate to the diagram in question. Feynman rule (f) says that each vertex of type B bearing site label  $i_j$  contributes the factor  $Q_{i_j}$ . Impurity averaging a typical graph with  $m$  vertices of type B, amounts to performing a sum of the form

$$\langle\langle Q_{i_1} \cdots Q_{i_m} \rangle\rangle = \sum_{\{p_i\}} P(\{p_i\}) Q_{i_1} \cdots Q_{i_m}. \quad (A1)$$

It follows from Eq. (7b) that, for completely distinct site labels  $i_1, i_2, i_3, \dots, i_m$

$$\langle\langle Q_{i_1} \cdots Q_{i_m} \rangle\rangle = 0 \quad (i_1 \neq i_2 \neq \cdots \neq i_m). \quad (A2)$$

This property is nothing more than the “site diagonal” nature of the randomness which has been built into the models we are considering.

In the event that some of the  $Q_i$ 's bear the same site label, the following result, which can be derived from Eqs. (1), (7b), and the fact that  $(p_i)^r = p_i$  for all positive integers  $r$  is useful:

$$(Q_i)^r = \alpha_r Q_i + \beta_r, \quad (A3)$$

where

$$\alpha_r = (1-p)^r - (-p)^r, \quad (A4)$$

$$\beta_r = p(1-p)^r + (1-p)(-p)^r. \quad (A5)$$

Relations (A2) and (A3) provide the most general prescription for averaging a product of  $Q_i$ 's:

$$\langle\langle (Q_{i_1})^{r_1} \cdots (Q_{i_m})^{r_m} \rangle\rangle = \beta_{r_1} \cdots \beta_{r_m} \quad (i_1 \neq i_2 \neq \cdots \neq i_m). \quad (A6)$$

Of course  $\beta_1 = 0$ , so that unless *each* vertex label appears more than once in the product of  $Q_i$ 's, the impurity average will give identically zero. This rule can be incorporated into our diagrammatic representation as follows<sup>21</sup>: (i) Take each graph arising from rule (a). Group the vertices of type B together in all possible distinct ways by joining them with dotted lines. Each different way of grouping gives rise to a separate diagram. The groups may be arbitrarily large, but each group must consist of at least two vertices. Thus there are no unpaired vertices of type B. (ii) Give each group of vertices one site label. Give each pair of lines emerging from a vertex a component label. (iii) With each solid line joining two vertices with site labels  $i$  and  $j$ , associate the factor  $L_{ij}^{-1}$ . Propagators are diagonal in the component indices. (iv) With each vertex of order 2S and type A, as-

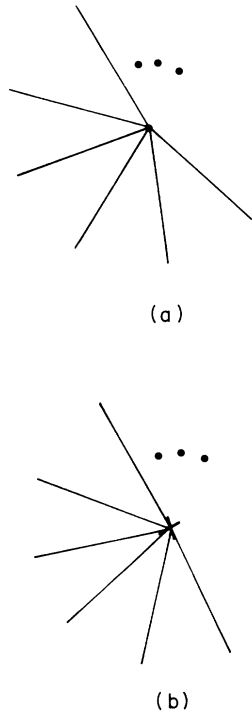


FIG. 4. (a) Vertex of type A and order 2S (2S lines emerge). (b) Vertex of type B and order 2S (2S lines emerge).

sociate the factor  $pC_{2S}$ . With each vertex of order  $2S$  and type  $B$ , associate the factor  $C_{2S}$ . (v) With each group containing  $r$  vertices of type  $B$ , associate the factor  $\beta_r$ . (vi) For a diagram with  $m$  distinct groups labeled  $(i_1, i_2, \dots, i_m)$  include the product

$$\prod_{i>k=1}^m (1 - \delta_{i_k, i_l}).$$

(vii) Multiply all factors arising from rules (iii)-(vi) together, and sum the product over all independent vertex labels and all independent component labels. (viii) Multiply the result of (vii) by a numerical factor appropriate to the diagram in question.

#### APPENDIX B: COMPLETE EFFECTIVE HAMILTONIAN

In Sec. III, we computed the lowest-order term of the effective Hamiltonian by graphically converting a dotted "impurity-averaging" line to a wavy interaction line. This same process can be applied to any graph of the impurity-averaged theory. If we interpret each group of vertices joined by wavy lines as a single interaction term of  $\mathcal{H}$ , we will build an infinite sequence of complex vertices for the effective Hamiltonian theory. A few examples of such vertices are shown in Fig. 5.

To calculate the coefficient which must be attached to a general interaction in  $\mathcal{H}$ , consider a complex vertex composed of  $n_s$  type  $B$  vertices of order  $2S$ , for  $S=1, 2, 3, \dots$ . Such a vertex will be denoted  $\{n_s\}$ . [For example, the vertex shown in Fig. 5(d) has  $n_1=1$ ,  $n_2=1$ , and all other  $n_i=0$ .] Feynman rules (iv) and (v) indicate that the coefficient

$$\beta_K \prod_{s=1}^{\infty} (C_{2S})^{n_s}$$

should be attached to this vertex. The number  $K$  is defined by

$$K = \sum_{s=1}^{\infty} n_s. \quad (\text{B1})$$

The combinatoric factor

$$\left( \prod_{s=1}^{\infty} \frac{1}{n_s!} \right)$$

is needed as well, to keep the numerical coefficients of the two theories identical.

Next we must deal with Feynman rule (vi), which calls for products of Kronecker  $\delta$ 's to appear in the analytic expressions corresponding to disorder-averaged graphs. An example of the effect of this rule will illustrate how it is handled in the general case. Consider the four graphs,  $G_1$  to  $G_4$  [Figs. 6(a)-6(d), respectively].

Because of Feynman rule (vi), the analytic expressions corresponding to  $G_1$ ,  $G_2$ , and  $G_3$  are each composed of two terms:

$$G_1 = (2C_2)^4 \beta_2^2 \left( \sum_{kp} L_{ik}^{-1} L_{kk}^{-1} L_{kp}^{-1} L_{pj}^{-1} - \sum_k L_{ik}^{-1} (L_{kk}^{-1})^3 L_{kj}^{-1} \right), \quad (\text{B2a})$$

$$G_2 = (2C_2)^4 \beta_2^2 \left( \sum_{kp} L_{ik}^{-1} L_{kp}^{-1} L_{kp}^{-1} L_{pj}^{-1} - \sum_k L_{ik}^{-1} (L_{kk}^{-1})^3 L_{kj}^{-1} \right), \quad (\text{B2b})$$

$$G_3 = (2C_2)^4 \beta_2^2 \left( \sum_{kp} L_{ik}^{-1} L_{kp}^{-1} L_{pp}^{-1} L_{pk}^{-1} L_{kj}^{-1} - \sum_k L_{ik}^{-1} (L_{kk}^{-1})^3 L_{kj}^{-1} \right), \quad (\text{B2c})$$

$$G_4 = (2C_2)^4 \beta_4 \left( \sum_k L_{ik}^{-1} (L_{kk}^{-1})^3 L_{kj}^{-1} \right). \quad (\text{B2d})$$

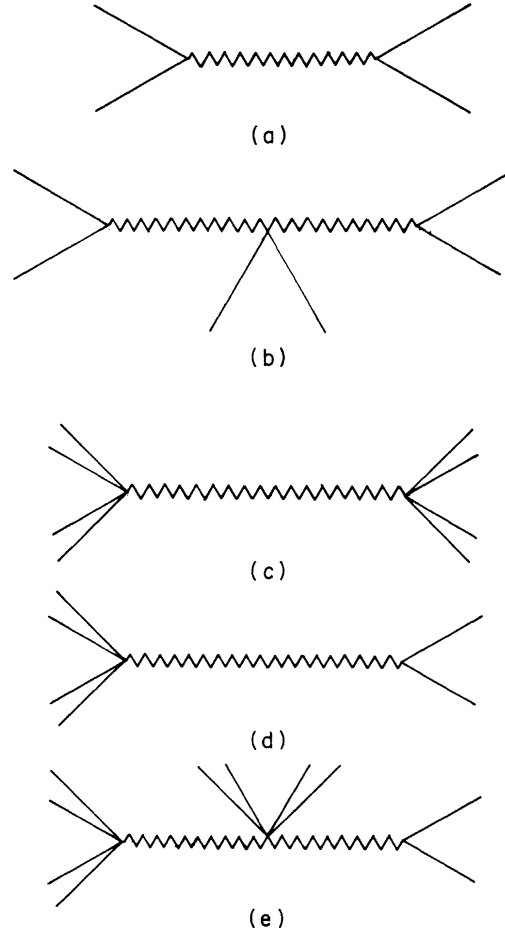


FIG. 5. Some examples of complex vertices occurring in the effective Hamiltonian theory.

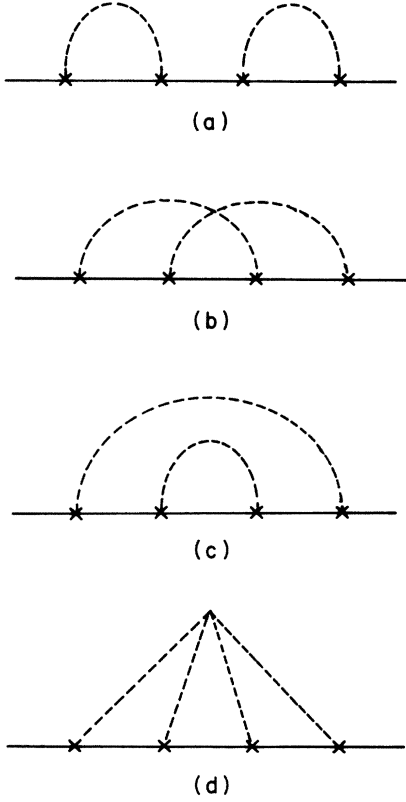


FIG. 6. Four graphs contributing to the "conventional" impurity-averaged theory.

It is clear that the corresponding  $\mathcal{H}$  graphs, shown in Figs. 7(a)–7(c), only compensate for the first terms of  $G_1$ – $G_3$ , respectively. The  $\mathcal{H}$  theory graph shown in Fig. 7(d) must account not only for  $G_4$ , but for the second terms of  $G_1$ – $G_3$  as well. We conclude that the coefficient of the  $\phi^8$  vertex which occurs in Fig. 7(d) must be  $(1/4!)C_2^4(\beta_4 - 3\beta_2^2)$ , and not simply  $(1/4!)C_2^4\beta_4$  as our previous naive analy-

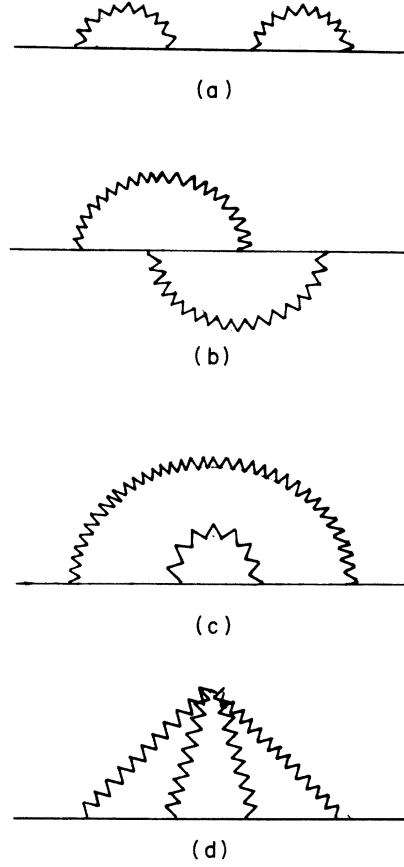


FIG. 7. Four graphs of the effective Hamiltonian theory corresponding to the "conventional" theory graphs of Fig. 6.

sis indicated. This type of difficulty occurs consistently in higher orders of perturbation theory. It can be resolved by everywhere replacing the function  $\beta_K$  with the function  $A_K$ , defined by

$$A_K = K! \sum_{m=1}^{\infty} (-1)^{m-1} (m-1)! \sum_{\substack{i=2,3,4,\dots \\ m_i=0,1,2,\dots}} \left[ \delta \left( \sum_i m_i - m \right) \delta \left( \sum_i (im_i) - K \right) \prod_{j=2}^{\infty} \frac{1}{m_j!} \left( \frac{\beta_j}{j!} \right)^{m_j} \right]. \quad (\text{B3})$$

That is, the coefficient of the general vertex  $\{n_s\}$  is

$$A_K \prod_{r=1}^{\infty} [(C_{2r})^{n_r} / (n_r!)],$$

where  $K$  is defined in Eq. (B1).

Finally, we must prove that overall combinatoric factors for graphs in the  $\mathcal{H}$  theory are identical to those of the conventional impurity-averaged theory described in Appendix A. The averaging process

for a graph in the latter theory leaves all combinatoric factors unchanged. Consequently, the combinatoric factor of such a graph does not depend on the dotted line structure at all. Consider any such graph [for example, the graph in Fig. 8(b)]. This graph arose from the diagram of Fig. 8(a), and produces the  $\mathcal{H}$  graph of Fig. 8(c). The numerical constant which multiplies the graph of Fig. 8(c) is greater than the constant multiplying Fig. 8(a) by a factor  $R$ , where  $R$  represents the

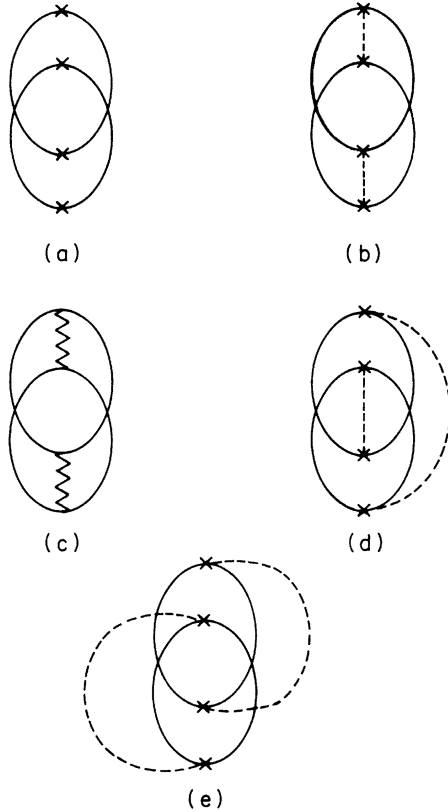


FIG. 8. (a) Graph contributing to the "conventional" theory prior to impurity averaging. (b) An impurity-averaged graph of the "conventional" theory. (c) The corresponding graph of the effective Hamiltonian theory. (d) and (e) Two graphs of the "conventional" theory which are topologically equivalent to Fig. 8(b).

number of different ways in which the dotted lines can be attached to the points in Fig. 8(a) to produce a diagram topologically equivalent to Fig. 8(b). [For example, the graphs of Figs. 8(d) and 8(e) are topologically equivalent to 8(b); hence  $R = 3$  in this case.] Feynman rule (i) requires us to count each of these  $R$  equivalent diagrams separately in the conventional theory, whereas they are treated as a single diagram in the  $\mathcal{H}$  theory. Therefore, the over-all numerical constants multiplying sets of corresponding diagrams in the two theories are the same. The perturbation expansions of the conventional and effective Hamiltonian theories are identical, term by term. The complete and unabridged effective Hamiltonian can be written

$$\begin{aligned} \mathcal{H}(\{\phi_{i\beta}^\alpha\}) = & \frac{1}{2} \sum_{i,j} \sum_{\alpha,\beta} L_{ij} \phi_{i\beta}^\alpha \phi_{j\beta}^\alpha \\ & + \sum_{s=2}^{\infty} p C_{2s} \sum_i \sum_{\beta=1}^n \left( \sum_{\alpha=1}^m (\phi_{i\beta}^\alpha)^2 \right)^s \\ & + \sum_{\substack{i \\ (\sum_j n_j > 1)}} \sum_{n_1, n_2, n_3, \dots=0}^{\infty} \left( A_K \prod_{s=1}^{\infty} \left\{ \frac{1}{n_s!} (C_{2s})^{n_s} \right. \right. \\ & \quad \left. \left. \times \left[ \sum_{\beta=1}^n \left( \sum_{\alpha=1}^m (\phi_{i\beta}^\alpha)^2 \right)^s \right]^{n_s} \right\} \right). \end{aligned} \quad (B4)$$

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- <sup>18</sup>D. E. Khmel'nitsky, Zh. Eksp. Teor. Fiz. 68, 1960 (1975).
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